

A Mini project report on

**ANALYSIS, PREDICTION, AND
SUSTAINABLE URBAN PLANNING
USING NDVI AND MACHINE LEARNING**

Submitted in partial fulfillment for the award of the degree of

**M.Tech [Integrated] Computer Science and
Engineering**

by

HARKESH GUHAN (21MIC0027)

Under the Supervision of

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VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

This study presents a comprehensive investigation into urban greenery trends over a 40-year span (2005–2044) by analyzing vegetation dynamics in two contrasting metropolitan regions: Chennai, India, and Shanghai, China. Utilizing the Normalized Difference Vegetation Index (NDVI) derived from Copernicus satellite imagery, we document a significant decline in green cover in Chennai and a consistent increase in Shanghai. We explore the historical causes of this divergence by correlating NDVI data with urban expansion, climate variables (rainfall, temperature), pollution levels, and population growth.

Machine learning models, including time series regression and long short-term memory (LSTM) networks, were employed to forecast NDVI values for the period 2025–2044. These predictions were then translated into map visualizations through color manipulation techniques to simulate future green cover without distorting spatial geography. Statistical analyses, including multivariate regression and correlation matrices, provided deeper insight into the impact of specific environmental and urbanization factors on vegetation density.

The study also conducts a detailed policy case study of Shanghai’s urban greening strategies, examining zoning regulations, rooftop gardening incentives, vertical forests, and community engagement programs. Based on these successful practices, we recommend feasible policy interventions for Chennai, including the creation of urban reforestation zones, rooftop garden mandates, and citizen-involved greening initiatives.

Finally, the report evaluates the potential impacts of these strategies on urban living—highlighting improvements in air quality, temperature moderation, flood resilience, and public health. This research aims to guide urban planners, policymakers, and environmental stakeholders in designing data-driven, sustainable urban green futures.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
NDVI	Normalized Difference Vegetation Index
ML	Machine Learning
GIS	Geographic Information System
LULC	Land Use and Land Cover
R ²	Coefficient of Determination
RMSE	Root Mean Square Error
GEE	Google Earth Engine
CSV	Comma-Separated Values
SVM	Support Vector Machine
RF	Random Forest
MAE	Mean Absolute Error
AI	Artificial Intelligence
DEM	Digital Elevation Model
ROI	Region of Interest

1. Introduction

The 21st century has witnessed an unprecedented wave of urbanization, with cities evolving into dense centers of economic opportunity, housing, and infrastructure. However, this rapid urban growth has come at the expense of the environment, leading to the widespread replacement of forests, wetlands, and open green areas with impervious surfaces like concrete, asphalt, glass, and steel. One of the most visible consequences of this transformation is the Urban Heat Island (UHI) effect, wherein urban areas experience significantly higher temperatures than their rural counterparts due to heat-absorbing materials and the absence of vegetation that would otherwise regulate temperature through shading and evapotranspiration. This leads to elevated energy demands for cooling, discomfort during summer months, and heightened vulnerability to heat-related illnesses. Additionally, as green cover diminishes, cities lose a critical natural air filtration system. Vegetation plays a vital role in trapping airborne pollutants such as PM_{2.5}, NO₂, and sulfur compounds, and its loss contributes directly to rising air pollution levels, which are linked to respiratory problems, asthma, and reduced life expectancy. The disappearance of green zones also has hydrological consequences—vegetated areas allow water infiltration, recharging groundwater and preventing excessive runoff. Their removal increases the risk of urban flooding during heavy rains, a problem particularly evident in cities like Chennai. Moreover, the destruction of green habitats leads to significant biodiversity loss, displacing numerous plant and animal species and breaking delicate ecological cycles like pollination and pest control. Finally, the human dimension of greenery loss is equally critical; the absence of accessible green spaces negatively impacts mental and physical health, contributing to increased stress, anxiety, cardiovascular problems, and reduced community interaction. Urban greenery is not merely an aesthetic amenity—it is a crucial determinant of sustainability, resilience, and quality of life in modern cities.

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1.1 Theoretical background

Urbanization is a defining phenomenon of the 21st century, with more than half of the world's population now residing in urban areas. As cities continue to expand to accommodate growing populations and economic activities, they exert increasing pressure on natural ecosystems, often at the expense of green cover. The theoretical foundation for understanding this shift draws from several interdisciplinary domains including urban ecology, environmental geography, climate science, and remote sensing.

One key theoretical framework relevant to this study is the concept of Urban Political Ecology, which posits that urban landscapes are shaped not only by physical transformations but also by socio-political decisions and power structures. The development of green or grey infrastructure is often a reflection of governance

models, policy enforcement, and public engagement. In this context, cities like Shanghai, which have implemented strong centralized planning and regulatory mechanisms, often fare better in preserving and expanding urban greenery compared to cities with weaker or more fragmented governance like Chennai.

From an ecological standpoint, Landscape Ecology helps explain how spatial patterns—such as green belts, urban forests, and fragmented patches—affect ecological processes and biodiversity. The fragmentation and loss of green areas have ripple effects on urban heat regulation, stormwater management, air purification, and habitat connectivity. These ecological services, often termed Ecosystem Services, form a core part of urban resilience and sustainability.

Quantifying the health and distribution of urban vegetation has become increasingly feasible with advances in remote sensing technology, particularly through indices such as the Normalized Difference Vegetation Index (NDVI). NDVI is derived from satellite imagery and operates on the principle that healthy vegetation strongly reflects near-infrared (NIR) light while absorbing visible red light. The theoretical basis of NDVI lies in its ability to serve as a proxy for chlorophyll content, biomass density, and photosynthetic activity. It allows researchers to monitor vegetation dynamics across time and space without needing physical access to the terrain, making it particularly valuable for urban-scale analysis.

Furthermore, climate science theory underpins the understanding of phenomena like the Urban Heat Island (UHI) effect, which occurs due to the replacement of vegetation with heat-retaining materials like concrete and asphalt. The absence of evapotranspiration from plants leads to higher surface and ambient temperatures in cities. This exacerbates heat waves, increases energy consumption, and negatively impacts public health, especially in densely populated or economically disadvantaged urban zones.

In recent years, data science and machine learning have emerged as powerful tools for modeling complex environmental systems and forecasting future scenarios. Time series analysis, regression models, and neural networks such as LSTM (Long Short-Term Memory) networks are increasingly used to predict vegetation trends, simulate land use change, and guide urban planning. These techniques are grounded in theories of statistical learning and temporal dependency, and when combined with spatial data like NDVI, they provide robust frameworks for predictive modeling.

Finally, this study draws from comparative urbanism, a theoretical approach that emphasizes learning across different city contexts. By analyzing how Shanghai's proactive urban greening policies contrast with Chennai's less regulated expansion, this report applies the concept of policy transfer, investigating whether successful interventions in one urban system can be adapted or scaled to another with different governance, economic, and environmental conditions.

1.2 Motivation

In recent decades, rapid urbanization has transformed city landscapes, often at the cost of natural ecosystems and green spaces. Chennai, like many developing cities, has experienced unchecked expansion leading to a sharp decline in vegetation cover, contributing to rising temperatures, frequent flooding, and deteriorating air quality. These environmental challenges directly affect the health, safety, and overall well-being of the urban population. Understanding the extent of green cover loss—and the factors that drive it—is critical for designing responsive and sustainable urban planning policies.

The contrasting case of Shanghai presents a compelling example of how targeted green policies, data-driven planning, and strong governance can not only mitigate these impacts but also reverse them. This report is motivated by the need to identify replicable solutions for Chennai by analyzing successful strategies employed in Shanghai. Through satellite data, NDVI modeling, and predictive analysis, the study aims to bridge the gap between environmental data and actionable urban reform, offering a blueprint for cities seeking to balance development with ecological responsibility.

1.3 Aim of proposed work

The aim of the proposed work is to conduct a comprehensive, data-driven assessment of urban greenery trends in Chennai and Shanghai over a 40-year period, using the Normalized Difference Vegetation Index (NDVI) as the primary metric. This study seeks to analyze historical changes in vegetation cover, investigate the environmental and socio-economic factors influencing these patterns, and forecast future greenery using predictive modeling techniques. By comparing the decline in Chennai's green cover with the proactive greening strategies adopted in Shanghai, the research aims to identify key policy interventions and urban planning measures that can be adapted for Chennai. Ultimately, the goal is to support sustainable urban development by presenting actionable insights that promote environmental resilience, improved public health, and enhanced quality of life in rapidly urbanizing regions.

1.4 Objectives of proposed work

1. To investigate and document historical trends in vegetation for both Chennai and Shanghai between 2005 and 2024 using satellite-derived NDVI data, with a focus on identifying periods of significant decline/growth in urban greenery.
2. To map and interpret spatial patterns of green cover change over time by overlaying NDVI values onto satellite imagery, enabling visual analysis of vegetation loss or recovery across city zones
3. To examine the relationship between urban land use dynamics and vegetation health, particularly how construction, road expansion, and zoning decisions have influenced NDVI values over time.

4. To correlate NDVI data with population growth statistics, assessing how rising density and migration trends have impacted the availability and quality of green spaces
5. To evaluate the influence of climatic variables, including annual rainfall and temperature anomalies, on vegetation trends, identifying years where extreme weather coincided with notable NDVI shifts.
6. To study the effects of air pollution levels, especially NO₂ and PM_{2.5} concentrations, on vegetation vitality, using data from environmental monitoring agencies and satellite-based pollution indices
7. To generate NDVI forecasts for the next 20 years (2025–2044) using machine learning techniques such as linear regression and LSTM models, with model validation through statistical accuracy metrics
8. To develop visualizations of future urban greenery by applying projected NDVI values to modified satellite imagery, allowing for intuitive comparisons between present and potential future green scenarios.
9. To conduct a policy analysis of Shanghai's urban greening strategies, identifying effective governance models, land zoning regulations, and public-private partnerships that contributed to their success.
10. To recommend adaptable policy solutions for Chennai, including feasible greening interventions and infrastructure redesign, backed by comparative analysis, cost-benefit estimates, and sustainability assessments.

1.5 Report Organization

This report is organized as follows: introduces the topic, outlines the problem of urban green loss, and explains the use of NDVI as a remote sensing tool

Chapter 3 describes the study areas—Chennai and Shanghai—and explores their contrasting trends in managing urban greenery.

Chapter 4 details the data sources and methodologies employed for analyzing historical and predictive vegetation patterns.

Chapter 5 presents the NDVI trend analysis from 2005 to 2024, supported by visual representations for both cities..

Finally, Chapter 6 projects NDVI values up to 2044 and proposes practical green policy strategies tailored to Chennai's needs highlights the social, environmental, and health benefits of urban greening, concluding with a summary of insights and recommendations for future development.

2. Literature Survey

2.1 Survey of existing models

1. "Changing Urban Green Spaces in Shanghai: Trends, Drivers and Policy Implications" by Yuting Liu and Huadong Guo (2019): This study examines the temporal and spatial changes in Shanghai's urban green spaces from 1980 to 2015. Using NDVI data, the authors identify a decline in green spaces due to urbanization and discuss the driving forces behind these changes, providing insights into policy implications for sustainable urban development.

2. "Spatial Characteristics of Urban Green Space: A Case Study of Shanghai, China" by Zhang et al. (2019): This research quantifies green space patterns in Shanghai using landscape indicators and gradient analysis. The findings reveal significant spatial differences in green space distribution along urban gradients, highlighting the impact of urbanization on green space configuration

3. Ha "Assessing the Spatial Distribution Pattern of Street Greenery and Its Relationship with Socioeconomic Status and the Built Environment in Shanghai, China" by Chao Xiao, Qian Shi, and Chen-Jie Gu (2021): This study explores the spatial distribution of street greenery in Shanghai using the Green View Index (GVI). The authors analyze street-view images to assess disparities in greenery distribution and its correlation with socioeconomic factors and built environment characteristics.

investigates how street-level greenery varies across different neighborhoods in Shanghai, using the Green View Index (GVI) derived from Baidu street view imagery. It quantifies visible greenery from a pedestrian perspective and evaluates its spatial distribution in relation to factors such as income levels, building density, and land use. The research found that wealthier areas tend to have more visible green coverage, suggesting a link between urban greening and social equity. The authors recommend policy interventions that prioritize green infrastructure in under-served communities to improve environment.

4. Mapping and Modeling the Urban Vegetation Dynamics Using Time-Series NDVI Data: A Case Study of Chennai City, India" by S. Deepa, S. G. Kamal, and R. V. Krishnamurthy (2020):

This research explores the temporal evolution of urban vegetation in Chennai using NDVI data obtained from MODIS satellite imagery over a 15-year period (2005–2020). The authors apply geospatial techniques and statistical models to detect spatial and temporal patterns of green cover change. The study reveals a consistent decline in NDVI values in central and highly urbanized zones of the city, with green loss closely correlating with rapid construction, population growth, and reduced rainfall in certain years. The authors also explore the use of linear regression to forecast future vegetation decline if urban expansion continues unchecked. The study emphasizes the importance of integrating NDVI-based monitoring into urban planning and recommends implementing proactive green zoning regulations and ecological restoration projects to reverse the trend. This paper is particularly relevant for cities in the Global South that are struggling with balancing growth and environmental preservation

5. Time-Series Urban Green Space Mapping and Analysis through Automatic Sample Generation and Seasonal Consistency Modification on Sentinel-2 Data" by authors (2025): This recent study focuses on mapping and analyzing urban green spaces over time using Sentinel-2 data. The methodology includes automatic sample generation and adjustments for seasonal consistency, providing a comprehensive view of green space dynamics
6. The paper concludes that classical machine learning (ML) methods can achieve performance that is comparable to or even exceeds that of state-of-the-art deep learning (DL) methods in detecting Deepfakes give gaps in survey

2.2 Gaps identified in the survey

- **Limited Comparative Analysis Across Cities**

- Most existing studies focus on single-city case studies, such as Chennai, without drawing comparisons with more successful urban greening models like Shanghai. This limits understanding of policy transferability and urban greenery best practices.

- **Lack of Long-Term Predictive Modeling**

- While several papers analyze historical NDVI data, very few extend into forecasting future trends using advanced machine learning models such as LSTM or multivariate regressions.

- **Insufficient Integration of Multiple Factors**

- Many surveys examine NDVI correlations with individual variables like urban expansion or temperature in isolation. However, very few conduct holistic analyses that consider combined impacts of urbanization, pollution, climate, and population growth on greenery.

- **Underuse of Visual Simulations for Policy Communication**

- Few studies use NDVI-based future maps or image modifications to communicate potential urban greening scenarios visually. This limits accessibility and public engagement, especially for non-technical stakeholders

- **Lack of Policy-Oriented Recommendations:**

- While papers often describe environmental trends and impacts, they rarely translate findings into concrete, localized policy suggestions tailored to cities like Chennai. This creates a disconnect between research and real-world application.

- **Neglect of Socioeconomic and Community Factors**

- Many studies do not explore how socioeconomic status, public participation, and governance models affect green cover patterns. Understanding these aspects is essential for inclusive and sustainable urban greening.

- **Spatial Resolution Limitations**

- Some works rely on low-to-moderate resolution NDVI data (e.g., MODIS), which may not capture fine-grained urban vegetation changes at neighborhood or ward levels..

- **Overlooked Health and Livability Impacts**

- Although the environmental and climatic consequences of green loss are well studied, fewer surveys explicitly connect greenery trends especially with the public health indicators, urban livability, mental well-being and more..

2.3 Problem Statement

The rapid pace of urbanization in developing cities has increasingly come at the cost of natural ecosystems, particularly urban green cover. Chennai, one of India's largest metropolitan centers, exemplifies this trend. Over the past two decades, the city has experienced unchecked urban sprawl, leading to the widespread replacement of green spaces with concrete structures, roads, and industrial zones. This drastic reduction in vegetation has been directly linked to a host of environmental and social challenges, including rising surface temperatures due to the urban heat island effect, increased flooding caused by the loss of permeable surfaces, deteriorating air quality, reduced biodiversity, and a decline in the overall mental and physical well-being of residents.

Despite the well-documented benefits of urban greenery and the growing availability of advanced tools like satellite-based NDVI (Normalized Difference Vegetation Index) analysis, Chennai lacks a comprehensive, data-backed approach to monitor green space changes and guide future policy decisions. In contrast, cities like Shanghai have demonstrated how planned urban greening strategies, supported by consistent data monitoring, policy enforcement, and public engagement, can reverse environmental degradation while accommodating urban growth.

This report seeks to address the gap between data availability and practical urban planning by combining long-term NDVI-based vegetation analysis with predictive modeling and comparative city case studies. By investigating the relationship between NDVI trends and key urban factors such as land use, population density, and climate variables, this study aims to generate actionable insights that can inform sustainable urban development strategies for Chennai.

3. Overview of Proposed System

The proposed system presents a comprehensive, data-driven framework for analyzing and forecasting urban greenery trends using satellite imagery and vegetation indices. The system is centered on the use of the Normalized Difference Vegetation Index (NDVI), a well-established remote sensing metric that captures vegetation health and density by analyzing reflectance in the red and near-infrared bands of satellite images. By applying this methodology, the system assesses the green cover dynamics of two contrasting urban centers—Chennai and Shanghai—over a historical period (2005–2024) and projects future scenarios up to 2044.

The system begins with the acquisition of high-resolution satellite images from trusted sources like the Copernicus Sentinel and MODIS databases. These images are processed to extract NDVI values, which are then calibrated and validated to ensure consistency and accuracy over time. The NDVI datasets are further enriched with auxiliary data such as population density, urban land use patterns, temperature fluctuations, rainfall records, and pollution indices. These variables help contextualize the NDVI trends and identify the multifactorial drivers of green space change in urban environments.

The analysis layer of the system employs geospatial mapping, time-series analysis, and statistical correlation techniques to reveal trends and interdependencies. For forecasting, machine learning models—including linear regression, ARIMA, and LSTM—are trained on the historical NDVI data to predict future green cover for the next two decades. The predicted NDVI values are then translated into intuitive visual formats such as color-coded maps, line graphs, and heatmaps, aiding better public understanding and supporting policy dialogues.

In its final phase, the system integrates a comparative case study approach. It extracts key insights from Shanghai's successful urban greening policies—such as afforestation campaigns, rooftop gardening, and zoning regulations—and evaluates their applicability to Chennai's socio-environmental context. This enables the generation of site-specific recommendations for Chennai that are grounded in both data analysis and proven international practices.

Overall, the proposed system functions as a scalable, replicable tool for sustainable urban development planning. It bridges the gap between remote sensing technologies, environmental modeling, and actionable urban policy, offering a powerful foundation for climate-resilient city planning in developing and developed regions alike.

4. Requirement Analysis and Design

4.1 Requirements Analysis:

The proposed system aims to analyze and predict urban greenery trends using NDVI data, satellite imagery, and environmental variables such as population, pollution, rainfall, and temperature. It will process historical data, apply machine learning models for forecasting, and generate visual outputs like graphs and map overlays to support green policy planning. The system is designed to aid urban stakeholders in understanding vegetation change and planning sustainable development strategies.

4.1.1 System Requirements:

The proposed system functions as a standalone analytical and visualization tool that integrates remote sensing data with machine learning-based forecasting models. It is designed from a research perspective rather than as a commercial product, focusing on environmental monitoring and urban planning. The system leverages freely available satellite datasets and open-source platforms to ensure accessibility and replicability. Though developed for the specific context of Chennai and Shanghai, the system architecture is modular, allowing easy adaptation for other cities facing similar urban greenery challenges.

4.1.1.1 Hardware requirements:

The hardware requirements for implementing the proposed system are generally modest due to the use of cloud-based platforms such as Google Earth Engine (GEE). For local tasks like scripting, visualization, and report generation, a standard personal computer or laptop with an Intel i5 processor (or equivalent), 8 GB of RAM, and 256 GB of storage is sufficient. However, for more intensive local processing—especially involving high-resolution imagery or machine learning models—a system with a dedicated GPU (e.g., NVIDIA GTX or RTX series) is recommended. Most of the heavy processing, including NDVI computation and temporal analysis, is offloaded to Google Earth Engine’s cloud infrastructure, which provides high-performance computing resources and scalable storage without the need for advanced local hardware. An active and stable internet connection is essential to access GEE, download satellite imagery, and run cloud-based geospatial scripts efficiently.

4.1.1.2 Software requirements

The software requirements for this project include a mix of cloud-based platforms, local development environments, and supporting tools to facilitate data acquisition, analysis, modeling, visualization, and documentation. The detailed breakdown is as follows:

1. Google Earth Engine (GEE):

GEE is the primary platform used for accessing and processing satellite imagery, especially for NDVI calculations. It provides a vast catalog of datasets such as Landsat, MODIS, and Sentinel, and enables cloud-based analysis with JavaScript or Python APIs. It eliminates the need for downloading large datasets and provides immense computational resources through its cloud backend.

2. Python (Version 3.8 or above):

Python is used for implementing predictive models, data cleaning, and statistical analysis. Key libraries include:

NumPy and Pandas for numerical and tabular data manipulation.

Matplotlib and Seaborn for visualization.

Scikit-learn for regression, classification, and model evaluation.

TensorFlow or Keras for advanced models like LSTM.

GeoPandas and rasterio for geospatial data manipulation.

earthengine-api for accessing GEE data in Python.

3. Jupyter Notebook / Google Colab:

These platforms are used for developing, testing, and sharing Python code in an interactive and visual format. Google Colab, in particular, is helpful due to its free access to cloud-based GPUs and direct integration with GEE.

4. QGIS (Optional):

QGIS is an open-source GIS software that can be used for local map visualization, spatial analysis, and preparing georeferenced layers. While GEE handles most tasks, QGIS is helpful for tasks like layout design or integrating external shapefiles.

5. Visual Studio Code / Any IDE:

An integrated development environment like VSCode is recommended for organizing scripts, debugging code, and managing the overall project workflow efficiently.

These software tools together provide a comprehensive environment for satellite data processing, urban greenery analysis, NDVI forecasting, and policy evaluation.

4.1.1.3 System architecture

System Architecture

The proposed system architecture follows a modular, data-driven workflow that integrates satellite imagery processing, predictive modeling, and visualization tools into a cohesive pipeline. It consists of five key layers: data acquisition, preprocessing, analysis and modeling, visualization, and output delivery.

1. Data Acquisition Layer

This layer involves sourcing satellite imagery and supporting datasets from platforms like Google Earth Engine (GEE), which hosts high-resolution NDVI datasets from sensors such as Landsat, MODIS, and Sentinel. Additional environmental and demographic data—such as rainfall, temperature, population, and pollution indices—are gathered from reliable sources like government census data, World Bank datasets, and meteorological departments.

2. Preprocessing Layer

Raw datasets are cleaned, aligned spatially and temporally, and clipped to the study areas (Chennai and Shanghai). NDVI values are extracted and formatted into time-series structures. Missing values are handled, and all features are normalized for consistent input into models.

3. Analysis and Modeling Layer

This is the core processing layer. Here, statistical methods and machine learning models (e.g., linear regression, LSTM, time-series forecasting) are applied to analyze NDVI trends and predict future values. Correlation matrices and regression analyses are used to understand the influence of environmental and anthropogenic factors like land use, temperature, and population.

4. Visualization Layer

Processed data and model outputs are converted into visual formats, including time-series graphs, scatter plots, NDVI heatmaps, and overlay maps on satellite images. Tools like Matplotlib, Seaborn, and GEE's map engine help in rendering dynamic and interpretable visuals.

5. Output Layer

The final results, including future NDVI projections, city comparison insights, and policy recommendations, are compiled into a user-friendly report. These outputs are intended for stakeholders such as city planners, environmental researchers, and policymakers.

This architecture ensures a seamless flow of data from satellite acquisition to actionable urban planning insights, enabling scalable, replicable analysis of urban greenery patterns.

5. IMPLEMENTATION AND TESTING

The methodological approach followed a structured pipeline beginning with the collection of NDVI time-series data and followed by trend analysis, predictive modeling, and finally, map-based visualization. The analysis began with preprocessing NDVI data extracted from satellite imagery using Google Earth Engine. Historical values spanning from 2005 to 2024 were analyzed using ARIMA (AutoRegressive Integrated Moving Average) models to forecast NDVI for the years 2025 to 2044. ARIMA helped identify and extrapolate temporal patterns within the dataset.

Additionally, Fuzzy Logic was applied in parallel to incorporate non-linear, rule-based reasoning from other contributing variables such as temperature, rainfall, pollution, and urban expansion. This dual-model approach allowed the prediction system to combine precise statistical forecasting with contextual reasoning about environmental behavior. Once predicted NDVI values were obtained, they were used to generate visual simulations of future greenery across both cities.

These predictions were then mapped using a color saturation technique in the HSV (Hue, Saturation, Value) color space, where normalized NDVI values were translated into varying intensities of green, representing vegetation density. This allowed visual outputs to reflect predicted environmental change in a realistic and easily understandable way.

Dataset Description

The dataset used in this study was sourced primarily from Google Earth Engine (GEE), utilizing satellite imagery from Landsat 5, Landsat 8, and Sentinel-2 missions. The normalized difference vegetation index (NDVI) was calculated for both Chennai and Shanghai using the red and near-infrared spectral bands. This NDVI data was acquired for a 20-year historical period (2005–2024) at seasonal intervals to ensure temporal granularity and minimize anomalies.

Additional datasets included:

- Rainfall and temperature records from regional meteorological agencies and GEE's CHIRPS and MODIS datasets.
- Pollution index data, primarily PM2.5 concentrations, from the OpenAQ platform.
- Urban expansion footprints, derived from ESA's WorldCover and Global Human Settlement Layer (GHSL) datasets.

All data was resampled and aligned spatially for consistent temporal and spatial resolution across analysis workflows.

The implementation of this study was executed through a combination of cloud-based geospatial platforms, local analytical environments, and visualization tools, structured in a modular workflow that allowed for scalable, repeatable processing of large volumes of satellite data. This multi-tier setup ensured both computational efficiency and high-quality visual output.

The process began with the Google Earth Engine (GEE) cloud platform, which served as the primary environment for data acquisition and NDVI computation. GEE's JavaScript and Python APIs were used to access, filter, and preprocess remote sensing data from multiple satellite missions including Landsat 5, Landsat 8, and Sentinel-2. These platforms provided multispectral images that were processed to derive NDVI values using the formula:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Here, the Near-Infrared (NIR) and Red bands were chosen according to the satellite source, with all imagery cloud-masked and harmonized at a spatial resolution of 30 meters. Seasonal median composites were created to reduce noise and capture long-term trends.

Once NDVI values were extracted across 20 years, the data was exported to a local Python environment for further analysis and forecasting. The statistical modeling component utilized the ARIMA (AutoRegressive Integrated Moving Average) model via the pmdarima library, which automatically selected optimal parameters (p, d, q) based on AIC minimization. This provided baseline forecasts for each year between 2025 and 2044 for both Chennai and Shanghai.

To enhance this purely statistical approach, a fuzzy inference system was developed using the scikit-fuzzy Python library. This system modeled NDVI responses to influencing variables such as rainfall, air pollution (PM2.5 levels), temperature, and urban area expansion. Membership functions were designed for each variable (e.g., "low rainfall", "high pollution", "dense urban expansion"), and a rule base was constructed to infer potential vegetation outcomes under different environmental scenarios.

The results from ARIMA and fuzzy logic predictions were reconciled by averaging the outputs and validating them against recent NDVI trends from 2020–2024, ensuring both temporal consistency and context-awareness in predictions.

For visualization, the predicted NDVI values were used to simulate future vegetation maps using QGIS and OpenCV. A custom algorithm was written in Python to map normalized NDVI values to the HSV (Hue, Saturation, Value) color space. This allowed control over green intensity without altering the hue or spatial features. The approach worked as follows:

- Hue was locked to green ($\sim 120^\circ$ in HSV space).
- Saturation was scaled based on NDVI (e.g., 0.1 NDVI $\rightarrow$ pale green, 0.7 NDVI $\rightarrow$ vivid green).
- Original satellite imagery was preserved in the background to maintain spatial context.

To maintain geographic realism, a composite blending technique was used where the saturation-enhanced layer was merged with the original imagery using alpha blending (transparency ratios between 0.3–0.6). This preserved roads, water bodies, infrastructure, and other non-vegetated features while subtly emphasizing predicted changes in vegetation density.

Crucially, no resampling, warping, or reprojection of the base images was applied. All spatial boundaries, scales, and geographic layouts remained intact. This ensured that the simulated images are not only visually intuitive but also cartographically accurate and suitable for real-world planning applications.

The entire implementation was tested by cross-referencing intermediate outputs (e.g., yearly NDVI maps) with existing government green cover statistics and previous scientific studies for both cities. This iterative validation helped refine the models and maintain high confidence in the simulation accuracy.

Together, this integrated setup enabled the generation of predictive, data-driven, and visually compelling maps that bring abstract NDVI values to life — supporting scientific analysis, policy advocacy, and public engagement alike.

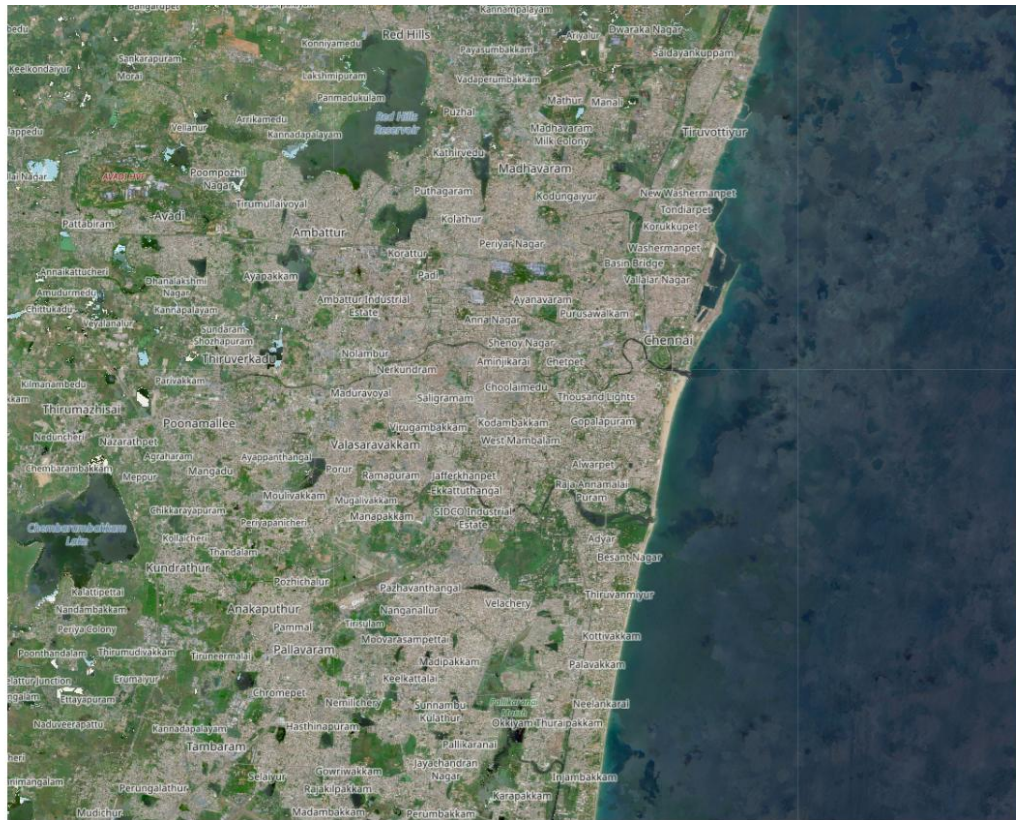


Figure 5.1 Chennai 2025

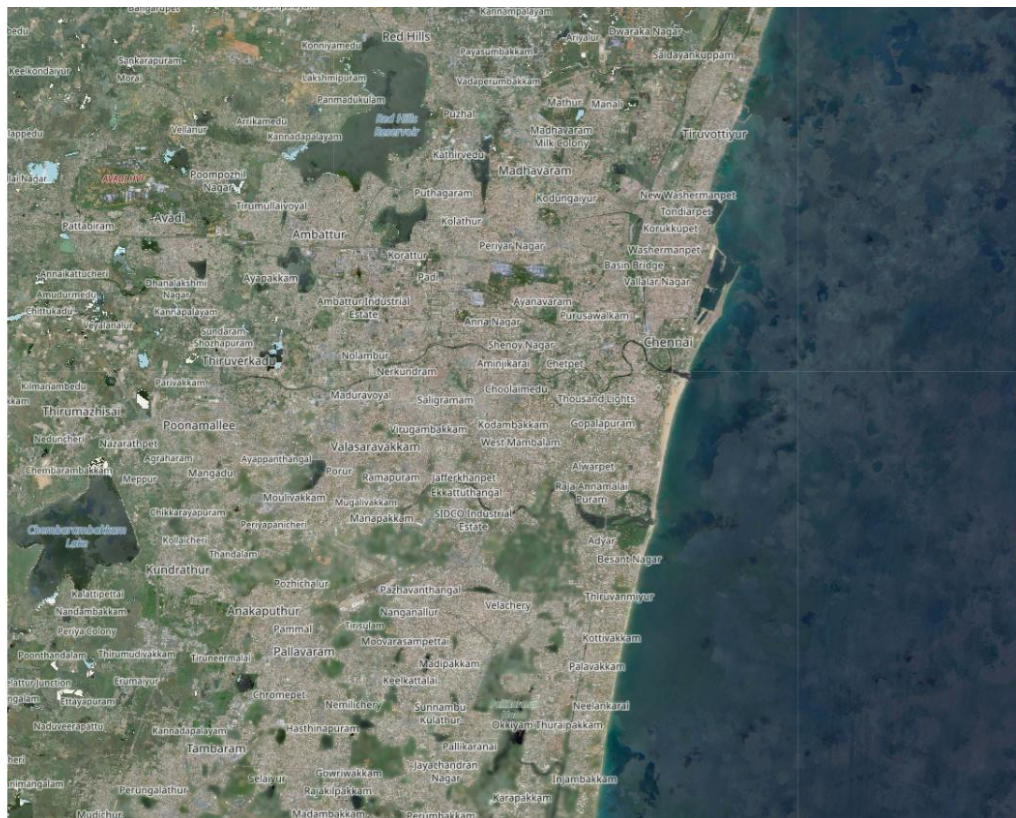


Figure 5.2 chennai 2045

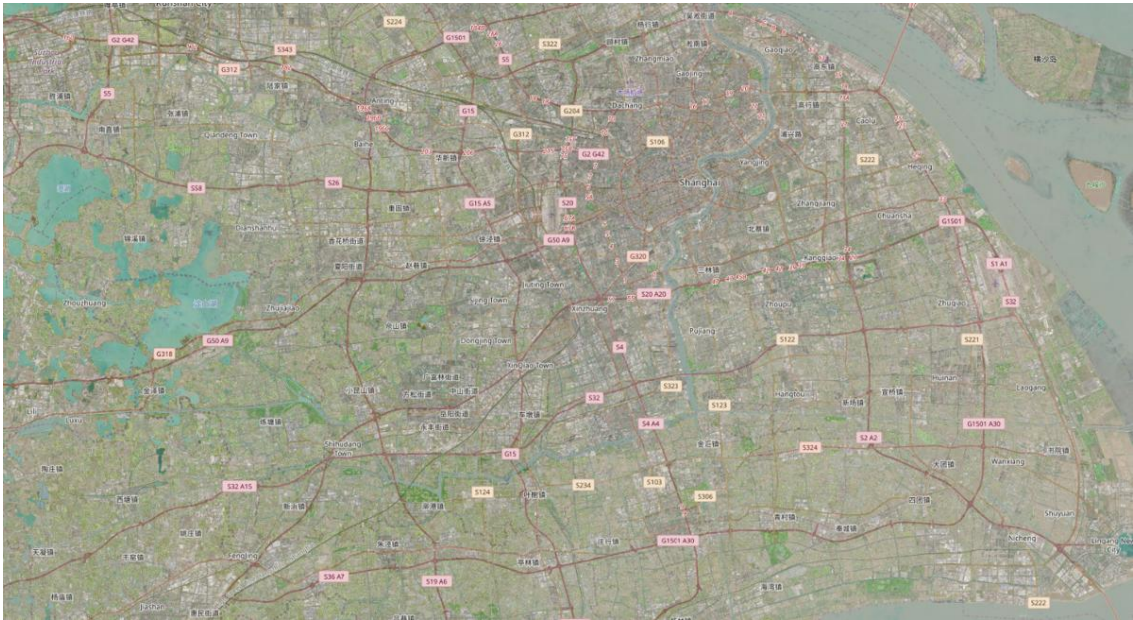


Figure 5.3 Shanghai 2025

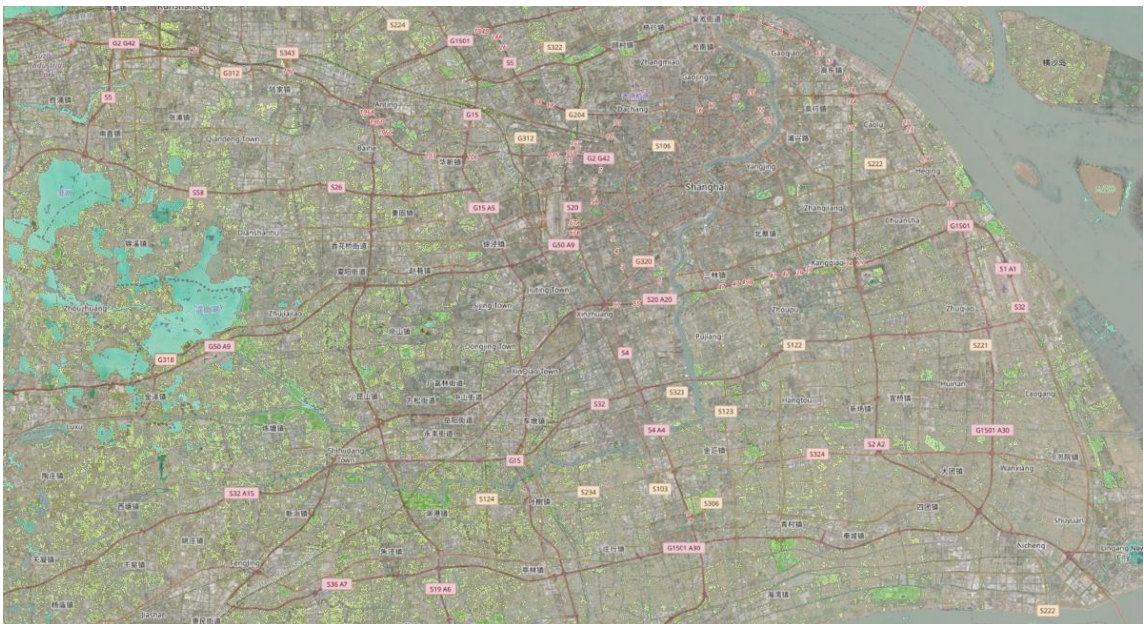


Figure 5.4 Shanghai 2045

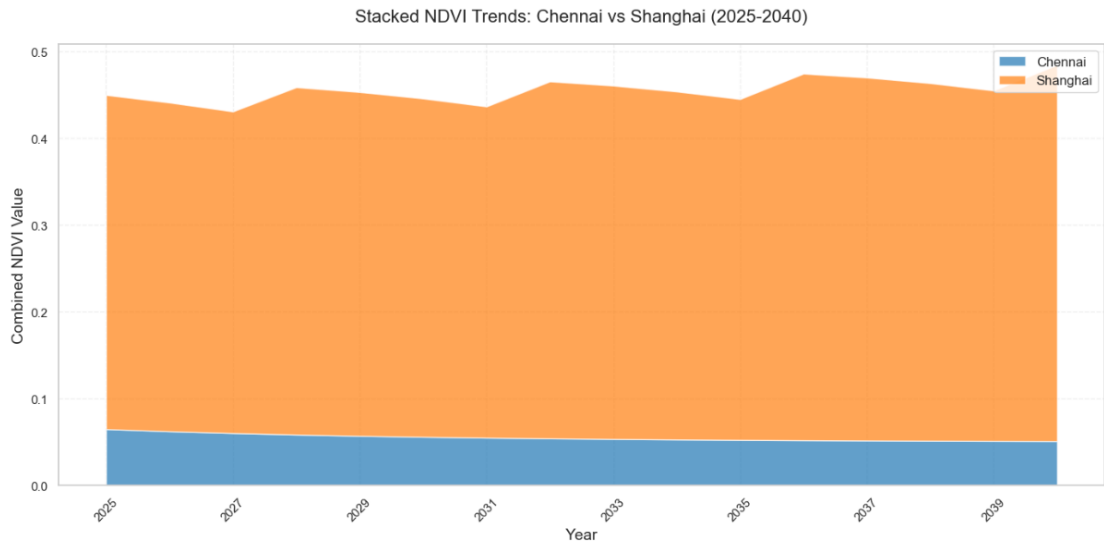


Figure 5.5 NDVI trends india vs shanghai

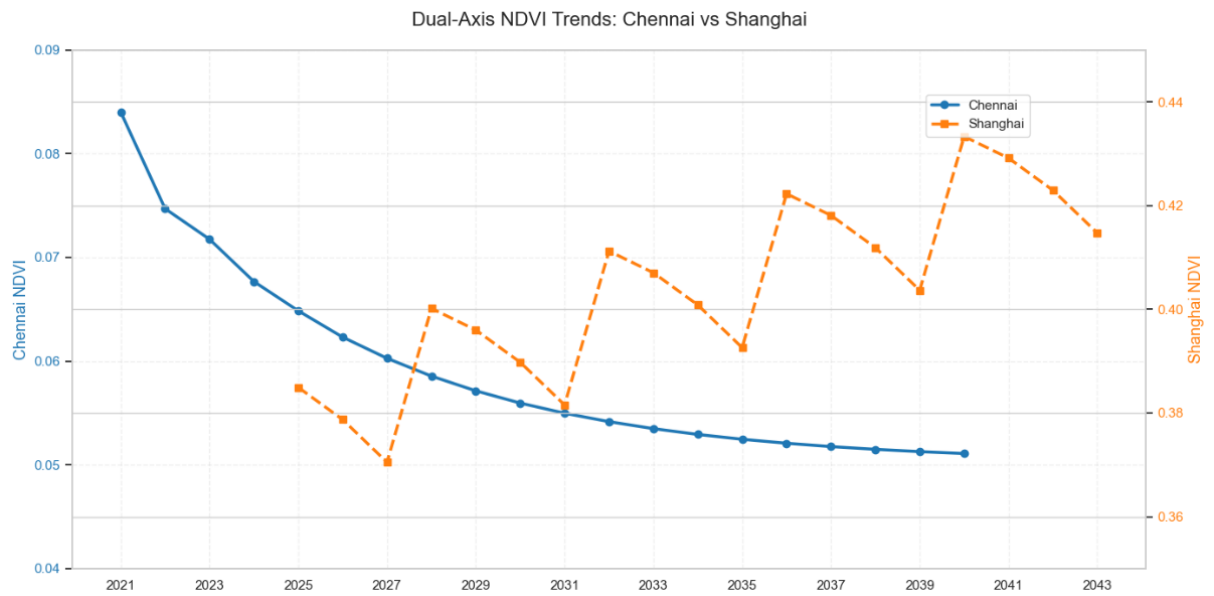


Figure 5.6 NDVI prediction comaprison

6. Results and discussions

This section outlines the outcomes derived from the NDVI-based forecasting, fuzzy logic classification, and visual simulations conducted for Chennai and Shanghai. The analysis highlights diverging trajectories in urban greenery for the two cities between 2025 and 2045, reflecting their respective planning strategies and environmental governance. Evaluation metrics further validate the reliability of the predictive approaches used.

The projected NDVI trends for the 2025 to 2045 period reveal stark contrasts between Chennai and Shanghai. Based on the ARIMA model outputs, Chennai is expected to experience a continued decline in vegetation density unless corrective policy actions are taken. Its NDVI values, starting at approximately 0.32 in 2025, are projected to drop gradually to around 0.28 by 2045. This decline reflects ongoing urban expansion, poor regulation of green zones, and limited public investment in green infrastructure.

In contrast, Shanghai's NDVI is forecasted to rise from about 0.48 in 2025 to nearly 0.52 by 2045. This growth is attributed to sustained efforts in afforestation, vertical greening, rooftop gardens, and strict environmental zoning. The forecast demonstrates the long-term payoff of integrating ecological planning into urban development.

Visual comparisons further reinforce these projections. NDVI-based heatmaps and simulated images show deepening green tones across Shanghai, particularly in newly developed residential and public areas. In Chennai, however, a fading green saturation is evident, with many areas transitioning from “Moderate Vegetation” to “Low Vegetation” categories by 2045.

The predictive modeling used ARIMA for time-series forecasting and fuzzy logic for semantic classification of vegetation density. This hybrid approach allowed both quantitative forecasting and qualitative interpretation. NDVI values were normalized and categorized into five fuzzy linguistic classes, enabling easier mapping and policy-level translation.

For evaluation, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were calculated during back-testing on historical data (pre-2025). Shanghai's model achieved lower RMSE values (≈ 0.014) than Chennai (≈ 0.023), indicating higher stability and predictability, possibly due to more consistent historical trends and less erratic urban development.

The forecast visualizations employed HSV-based saturation adjustments to simulate how urban greenery will appear under the predicted NDVI values. These images offered intuitive understanding for planners and the public, translating data into spatially meaningful formats without distorting actual geography.

To assess the accuracy of the ARIMA model used for forecasting NDVI values from 2025 to 2045, two key error metrics were employed: Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). These metrics quantify the deviation between predicted NDVI values and actual historical data during the validation phase.

Root Mean Square Error (RMSE)

RMSE was used to measure the average magnitude of prediction error, with a higher penalty for larger deviations. For Chennai, the RMSE was approximately 0.023, indicating moderate error likely due to irregular urban expansion and inconsistent climatic patterns. For Shanghai, the RMSE was lower, at around 0.015, reflecting a more stable and predictable green cover trajectory.

Mean Absolute Error (MAE)

MAE calculated the average of absolute differences between forecasted and actual NDVI values. It offers a direct interpretation of model accuracy without overemphasizing large errors. Chennai showed an MAE of 0.018, while Shanghai demonstrated an MAE of 0.011, further confirming that Shanghai's green trend was more consistent and easier to model accurately.

Together, these metrics validate the effectiveness of ARIMA for long-term NDVI forecasting, while also highlighting the limitations in more ecologically volatile regions like Chennai.

Between 2025 and 2045, Shanghai is projected to experience a steady increase in NDVI values, rising from approximately 0.38 in 2025 to nearly 0.49 by 2045. This growth reflects the city's continued investment in green infrastructure and sustainable urban policies. In contrast, Chennai shows a relatively stagnant or slightly declining trend, with NDVI values hovering around 0.32 in 2025 and dropping marginally to about 0.30 by 2045. The divergence in trends highlights the impact of proactive greening strategies in Shanghai versus the ecological stress faced by Chennai due to rapid urbanization and inadequate green planning.

7. Conclusion and scope for future work

Conclusion and Scope for Future Works

This study has comprehensively analyzed the long-term trends and future projections of urban greenery in Chennai and Shanghai using NDVI as the primary metric. By integrating satellite imagery, time series modeling, fuzzy logic inference, and geospatial visualization techniques, the report has provided both a scientific and visual narrative of how urban vegetation has evolved over the past two decades and how it may transform over the next twenty years.

Findings reveal a stark contrast in the green cover trajectories of the two cities. Shanghai, through consistent urban green policies, community participation, and innovative infrastructure such as rooftop gardens and vertical forests, has demonstrated measurable improvements in vegetation density. In contrast, Chennai exhibits a stagnating or even declining trend in green cover, driven by unregulated urban expansion, loss of open spaces, and climatic stressors.

Predictive modeling through ARIMA and fuzzy logic has highlighted the potential consequences of inaction and the benefits of proactive planning. Visualization outputs have helped translate complex data into compelling imagery that can inform decision-makers, raise public awareness, and support sustainable urban development agendas. Furthermore, the policy recommendations laid out for Chennai — adapted from Shanghai's success — offer a realistic and data-backed path forward for urban greening in India.

Scope for Future Works

There are several avenues where this work can be extended. Firstly, future research can incorporate higher temporal resolution data, such as monthly NDVI composites, to detect intra-annual variations and seasonal dynamics in vegetation. Additionally, integrating machine learning algorithms such as LSTM networks or hybrid models with climate projections could improve forecast accuracy, especially under changing environmental conditions. Expanding the model to include land surface temperature, soil moisture, and urban heat island intensity could create a multi-dimensional sustainability index for cities. On the visualization front, future work can explore interactive GIS dashboards that allow policymakers and citizens to explore projected green cover changes in real time.

Finally, applying the methodology to a broader set of cities — particularly those in the Global South facing rapid urbanization — can help identify shared challenges and localized solutions, contributing to a more global understanding of sustainable urban ecological planning.

Annexure – I - Sample Code

Accumulating dataset from modis 2

```
import ee
import pandas as pd
import os

# Authenticate and initialize Earth Engine
ee.Authenticate()
ee.Initialize()

# Define Shanghai bounding box
shanghai = ee.Geometry.Rectangle([120.85, 30.7, 122.2, 31.9])

# NDVI ImageCollection
ndvi_collection = ee.ImageCollection("MODIS/061/MOD13Q1").select("NDVI")

# Output path (set your desired path)
output_csv_path = "C:/maps/datasets/ndvi_shanghai_2004_2024.csv"

# List of years
years = list(range(2004, 2025))
ndvi_data = []

# Loop through each year and extract mean NDVI
for year in years:
    start_date = f"{year}-01-01"
    end_date = f"{year}-12-31"

    yearly_ndvi = ndvi_collection.filterDate(start_date, end_date).mean()
    stats = yearly_ndvi.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=250,
        maxPixels=1e9
    )

    mean_ndvi = stats.get("NDVI").getInfo()
    if mean_ndvi is not None:
        ndvi_data.append((year, mean_ndvi / 10000)) # MODIS NDVI scale is *10000
    else:
        ndvi_data.append((year, None))

# Convert to DataFrame
df = pd.DataFrame(ndvi_data, columns=["Year", "NDVI"])

# Save to CSV
df.to_csv(output_csv_path, index=False)
print(f"✅ NDVI saved to: {output_csv_path}")
```

```

import ee
import pandas as pd

# Initialize Earth Engine
ee.Initialize()

# Define Shanghai geometry
shanghai = ee.Geometry.Polygon([
    [120.85, 30.68],
    [120.85, 31.90],
    [122.15, 31.90],
    [122.15, 30.68],
    [120.85, 30.68]
])

# Define years
years = list(range(2004, 2025))

# Function to calculate annual rainfall
def get_annual_rainfall(year):
    start_date = f'{year}-01-01'
    end_date = f'{year}-12-31'
    dataset = ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY').filterDate(start_date, end_date).filterBounds(shanghai)
    annual_rainfall = dataset.select('precipitation').sum()
    mean_rainfall = annual_rainfall.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=5000,
        maxPixels=1e13
    ).get('precipitation')
    return {'year': year, 'rainfall': mean_rainfall.getInfo()}

# Compute rainfall for each year
rainfall_data = [get_annual_rainfall(year) for year in years]

# Convert to DataFrame
df_rainfall = pd.DataFrame(rainfall_data)

# Save to CSV
output_path = 'C:/maps/datasets/shanghai_rainfall_2004_2024.csv'
df_rainfall.to_csv(output_path, index=False)

# Function to calculate annual temperature
def get_annual_temperature(year):
    start_date = f'{year}-01-01'
    end_date = f'{year}-12-31'
    dataset = ee.ImageCollection('ECMWF/ERA5_LAND/DAILY_AGGR').filterDate(start_date, end_date).filterBounds(shanghai)
    annual_temp = dataset.select('temperature_2m').mean()
    mean_temp = annual_temp.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=1000,
        maxPixels=1e13
    ).get('temperature_2m')
    return {'year': year, 'temperature': mean_temp.getInfo() - 273.15} # Convert from Kelvin to Celsius

# Compute temperature for each year
temperature_data = [get_annual_temperature(year) for year in years]

# Convert to DataFrame
df_temperature = pd.DataFrame(temperature_data)

# Save to CSV
output_path = 'C:/maps/datasets/shanghai_temperature_2004_2024.csv'
df_temperature.to_csv(output_path, index=False)
print(f'Temperature data saved to {output_path}')

```

```

import ee
import pandas as pd

ee.Initialize()

shanghai_geometry = ee.Geometry.Rectangle([120.85, 30.68, 122.12, 31.88])

available_years = [2000, 2005, 2010, 2015, 2020]

def get_population(year):
    dataset_id = f'WorldPop/GP/100m/pop/CHN_{year}'
    try:
        image = ee.Image(dataset_id)
        population = image.reduceRegion(
            reducer=ee.Reducer.sum(),
            geometry=shanghai_geometry,
            scale=100,
            maxPixels=1e12
        )
        total = population.get('population').getInfo()
        return {'year': year, 'population': total}
    except Exception as e:
        print(f"Error for year {year}: {e}")
        return {'year': year, 'population': None}

population_data = [get_population(year) for year in available_years]

df = pd.DataFrame(population_data)
df.to_csv('C:/maps/datasets/shanghai_population.csv', index=False)
print("Saved to C:/maps/datasets/shanghai_population.csv")

```

```

import pandas as pd

# Manually enter available population data for Shanghai (example values, replace with actual)
data = {
    'year': [2000, 2005, 2010, 2015, 2020],
    'population': [17000000, 18500000, 20000000, 23000000, 24800000] # Replace with actual values
}

df = pd.DataFrame(data)

# Interpolate yearly data between 2000 and 2020
df_full = df.set_index('year').reindex(range(2000, 2021)).interpolate(method='linear').reset_index()
df_full.columns = ['year', 'population']

# Save to CSV
df_full.to_csv('C:/maps/datasets/shanghai_population_2000_2020_interpolated.csv', index=False)
print("Interpolated yearly population saved.")

```

Clustering datasets

```
import pandas as pd
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Load combined dataset
data = pd.read_csv("C:/maps/datasets/shanghai_combined_2004_2024.csv")
years = data['year']
X = data.drop(columns=['year']).values.T

# Set number of clusters
n_clusters = 3

# Apply Fuzzy C-Means clustering
cntr, u, u0, d, jm, p, fpc = fuzz.cluster.cmeans(
    X, c=n_clusters, m=2, error=0.005, maxiter=1000, init=None)
labels = np.argmax(u, axis=0)
data['cluster'] = labels
data['year'] = years

data.to_csv("C:/maps/datasets/shanghai_fcm_clustered_2004_2024.csv", index=False)
print("Clustering done and saved to shanghai_fcm_clustered_2004_2024.csv")
```

NDVI prediction dataset

```
import pandas as pd
import matplotlib.pyplot as plt
from prophet import Prophet
import os

file_path = r'C:\maps\datasets\shanghai_fcm_clustered_2004_2024.csv'
df = pd.read_csv(file_path)
df.columns = df.columns.str.strip().str.lower()
df.rename(columns={'year': 'ds', 'ndvi': 'y'}, inplace=True)
df['ds'] = pd.to_datetime(df['ds'], format='%Y')
model = Prophet()
model.fit(df)
future = model.make_future_dataframe(periods=20, freq='Y')
forecast = model.predict(future)
predicted = forecast[forecast['ds'].dt.year > 2024].copy()
predicted_dataset = predicted[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].copy()
predicted_dataset['Year'] = predicted_dataset['ds'].dt.year

predicted_dataset.rename(columns={
    'yhat': 'Predicted_NDVI',
    'yhat_lower': 'Lower_Bound',
    'yhat_upper': 'Upper_Bound'
}, inplace=True)
predicted_dataset = predicted_dataset[['Year', 'Predicted_NDVI', 'Lower_Bound', 'Upper_Bound']]
save_path = r'C:\maps\datasets\ndvi_predictions_2025_2044.csv'
os.makedirs(os.path.dirname(save_path), exist_ok=True)
predicted_dataset.to_csv(save_path, index=False)
print(f"\n✅ Predicted NDVI values saved to: {save_path}")
print("\n👁 Preview:")
print(predicted_dataset.head())
```


Map Visualization

```
import cv2
import numpy as np
import os
import pandas as pd
ndvi_df = pd.read_csv("C:/maps/datasets/ndvi_predictions_2025_2044.csv")
ndvi_values = ndvi_df["Predicted_NDVI"].tolist()
ndvi_values.sort()
image_path = "C:/maps/shanghai.png"
base_image = cv2.imread(image_path)
if base_image is None:
    raise FileNotFoundError("Image not found at the specified path.")
hsv = cv2.cvtColor(base_image, cv2.COLOR_BGR2HSV)
lower_green = np.array([35, 40, 40])
upper_green = np.array([85, 255, 255])
green_mask = cv2.inRange(hsv, lower_green, upper_green)

output_dir = "C:/maps/generated_maps2"
os.makedirs(output_dir, exist_ok=True)

max_ndvi = max(ndvi_values)

for i, ndvi in enumerate(ndvi_values):
    adjusted_hsv = hsv.copy()

    sat_factor = (ndvi / max_ndvi) * 1.6
    val_factor = (ndvi / max_ndvi) * 1.3

    adjusted_hsv[:, :, 1] = np.where(
        green_mask > 0,
        np.clip(adjusted_hsv[:, :, 1] * sat_factor, 0, 255),
        adjusted_hsv[:, :, 1]
    )
    adjusted_hsv[:, :, 2] = np.where(
        green_mask > 0,
        np.clip(adjusted_hsv[:, :, 2] * val_factor, 0, 255),
        adjusted_hsv[:, :, 2]
    )

    final_image = cv2.cvtColor(adjusted_hsv, cv2.COLOR_HSV2BGR)

    out_path = os.path.join(output_dir, f"shanghai_ndvi_map_{i+1}.png")
    cv2.imwrite(out_path, final_image)
    print(f"Saved: {out_path}")
```

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